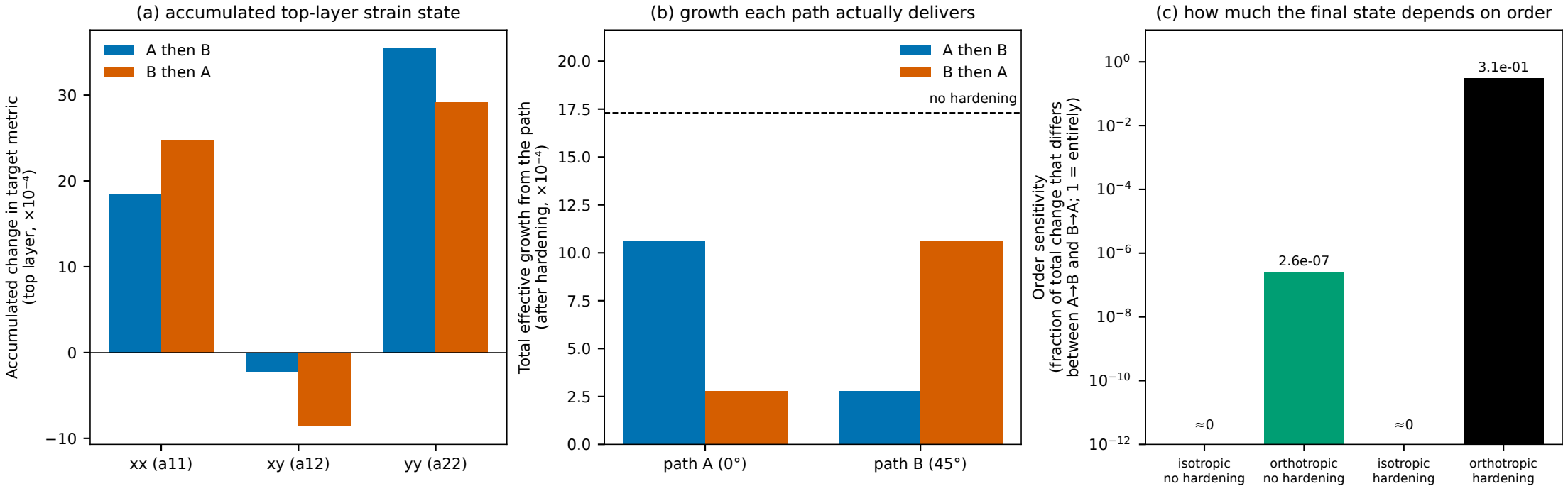
(a) accumulated top-layer strain state
A then B
B then A
Accumulated change in target metric
(top layer, ×10⁻⁴)
30
20
10
0
−10
xx (a11)
xy (a12)
yy (a22)
(b) growth each path actually delivers
A then B
B then A
no hardening
Total effective growth from the path
(after hardening, ×10⁻⁴)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
path A (0°)
path B (45°)
(c) how much the final state depends on order
Order sensitivity
(fraction of total change that differs
between A→B and B→A; 1 = entirely)
10⁰
10⁻²
10⁻⁴
10⁻⁶
10⁻⁸
10⁻¹⁰
10⁻¹²
≈0
2.6e-07
≈0
3.1e-01
isotropic
no hardening
orthotropic
no hardening
isotropic
hardening
orthotropic
hardening